Comprehensive Topography of YouTube and Google Analytics API Systems: A Metadata and Metrics Reference Architecture

The modern digital video ecosystem operates on a foundation of hyper-granular data telemetry. For data scientists, platform strategists, and performance analysts, understanding the precise mechanisms by which user behavior is quantified is paramount. The YouTube data infrastructure is not a singular monolithic entity; rather, it is a complex, multi-tiered architecture comprising the YouTube Data API v3, the YouTube Analytics API, and the YouTube Reporting API, further augmented by external integrations via Google Analytics 4 (GA4).

This report provides an exhaustive, expert-level deconstruction of every available data point, metric, and dimension across these systems. The data is categorized into core architectural dimensions, channel-wide statistics, universal individual video metrics, long-form-specific analytics, and short-form-specific analytics. Each data point is detailed with its API nomenclature, calculation methodology, differentiation from similar metrics, and second-order strategic implications.

1. Systemic Architecture of YouTube Data Interfaces

To parse the metrics accurately, the structural distinctness of the systems delivering them must be established. The YouTube API ecosystem separates metadata management from performance analytics, and further divides performance analytics based on the required query latency, data volume, and reporting latency.[1]

The first pillar is the **YouTube Data API v3**. This system manages the structural hierarchy of a channel. It handles metadata such as titles, descriptions, and publishing status, and provides basic, public-facing aggregate statistics via the statistics object.[4] It is primarily utilized for content management, application development, and workflow automation rather than deep behavioral analysis.[5]

The second pillar is the **YouTube Analytics API**. Designed for real-time, targeted queries, it allows systems to generate custom reports using dynamic combinations of metrics and dimensions.[1] It utilizes camelCase formatting for its data objects (e.g., averageViewDuration, trafficSourceType).[1] This API is heavily constrained by query quotas and is optimized for immediate, dashboard-level data retrieval.[1]

The third pillar is the **YouTube Reporting API**. Built for asynchronous, bulk data extraction, it generates daily, system-managed CSV-style reports where each row's unique combination of dimensions acts as a primary key.[2] This system uses snake_case formatting (e.g., average_view_duration_seconds, traffic_source_type) and is essential for data warehousing, machine learning integrations, and large-scale architectural ingestion into systems like Google BigQuery.[2]

Finally, **Google Analytics 4 (GA4)** serves as an external integration layer. While YouTube APIs track events occurring natively on the platform, GA4 integrations measure how embedded YouTube videos perform on external websites and applications. This utilizes event-based tracking schemas such as video_start, video_progress, and custom configured metrics like video_view.[11]

Across the YouTube Analytics and Reporting APIs, data fields are strictly categorized as either **Dimensions** or **Metrics**.[1] Dimensions are the criteria used to aggregate and group data (such as geography or device type), while Metrics are the quantitative measurements of user activity or revenue.[7] Certain metrics and dimensions are classified as "Core," meaning they are fundamental to the API and are subject to the official Google Deprecation Policy.[2] This policy guarantees that these specific data points will not be removed or structurally altered without substantial advance notice.[2] Non-core metrics can be modified, renamed, or removed at the platform's discretion.[1]

2. Core and Additional Dimensions: The Axes of Measurement

Metrics exist in a vacuum until they are intersected with Dimensions. Dimensions aggregate the metrics across time, geography, user demographic, and hardware, functioning as the primary keys in bulk data reports.[2]

Temporal and System Dimensions

Temporal and systemic dimensions establish the fundamental boundaries of any data query. They dictate the chronological window and the structural ownership of the media being analyzed.

Common Name	API Formatted Title	Description, Calculation, and Differentiation
Day	day / date	**Description:** The chronological grouping of data by specific calendar days. In the Reporting API, a date represents the period from 12:00 AM to 11:59 PM Pacific Time (UTC-7 or UTC-8).[1] **Differentiation:** It provides the highest level of temporal granularity in bulk reports. Unlike real-time Data API statistics, Analytics/Reporting API dates represent finalized, reconciled daily aggregates.[1]
Month	month	**Description:** Aggregates data by a specific calendar month.[1] **Differentiation:** Used exclusively in the Analytics API for broader trend analysis, reducing the total row count in payload responses compared to daily queries.[1]
Video ID	video / video_id	**Description:** The unique, 11-character alphanumeric identifier for a YouTube video.[1] **Differentiation:** Serves as the fundamental anchor for individual content performance. When included in a report alongside the channel dimension, it breaks down channel-wide aggregates into video-specific line items.[2]
Channel ID	channel / channel_id	**Description:** The unique identifier for a YouTube channel.[1] **Differentiation:** Frequently utilized in Content Owner reports because these reports typically aggregate data across multiple channels managed by a single network or entity.[1]
Playlist ID	playlist / playlist_id	**Description:** The unique identifier of a YouTube playlist.[1] **Differentiation:** Allows analysts to isolate metrics generated strictly within the sequential playback environment of a curated list, as opposed to organic standalone discovery.[2]
Uploader Type	uploaderType / uploader_type	**Description:** Utilized in Content Owner reports to distinguish between content uploaded directly by the owner (self) versus content uploaded by third parties (thirdParty).[1] **Differentiation:** This dimension is critical for copyright management and piracy tracking, enabling labels and studios to separate organic channel growth from revenue generated by claiming fan-uploaded material.[2]

Asset ID	asset_id (Reporting API)	**Description:** The unique identifier for an asset within YouTube's Content ID system, representing intellectual property such as a sound recording or a television episode.[14] **Differentiation:** Exists only in Content Owner reports. A single video_id might contain multiple asset_ids (e.g., a visual asset and an audio asset), allowing rights holders to track performance on a granular intellectual property level rather than just the video container level.[15]
Claimed Status	claimedStatus / claimed_status	**Description:** Indicates if the data row contains metrics for content claimed via Content ID. The only valid value populated is claimed.[1] **Differentiation:** Acts as a binary filter in enterprise reporting to isolate data rows that represent monetized copyright claims against user-generated content.[2]
Content Type	creatorContentType	**Description:** Identifies the structural format of the media, such as LIVE_STREAM, SHORTS, STORY, VIDEO_ON_DEMAND, or UNSPECIFIED.[1] **Differentiation:** Crucial for segmenting performance benchmarks. It prevents short-form data (which inherently possesses lower watch times and higher view velocities) from skewing the long-form VOD averages in mixed-format channels.[1]

The interaction between systemic dimensions dictates the scope of enterprise analytics. For large media conglomerates, record labels, and multi-channel networks, analyzing individual channels via the channel_id dimension is insufficient. The Reporting API relies heavily on the asset_id, uploader_type, and claimed_status dimensions to manage intellectual property.[15] By cross-referencing uploader_type set to thirdParty with a claimed_status of claimed, a music label can isolate the exact financial yield of secondary fan creations or pirated uploads that the Content ID system has successfully monetized on their behalf.

Audience Demographic Dimensions

Demographic dimensions allow analysts to construct precise user profiles, dictating advertising value and content alignment.

Common Name	API Formatted Title	Description, Calculation, and Differentiation
Age Group	ageGroup / age_group	**Description:** Categorizes logged-in viewers into standardized age brackets (e.g., AGE_13_17, AGE_25_34, AGE_65_). As of a policy update in March 2026, this dimension officially includes users estimated to be under 18.[2] **Differentiation:** Age directly dictates the CPM (Cost Per Mille) value of the audience. It filters out non-logged-in traffic, meaning the raw numbers will always be a subset of total platform reach.[1]
Gender	gender / gender	**Description:** The self-identified or algorithmically inferred gender of the logged-in user.[1] **Differentiation:** Often utilized alongside ageGroup to build composite viewer profiles for targeted brand sponsorships.[2]
Subscription Status	subscribedStatus / subscribed_status	**Description:** Indicates whether the user was SUBSCRIBED or UNSUBSCRIBED precisely at the moment the viewing activity occurred.[1] **Differentiation:** This dimension is dynamic. If a user watches a video while unsubscribed, subscribes mid-video, and watches a second video, the first view is logged as UNSUBSCRIBED and the second as SUBSCRIBED.[1] It separates core audience retention from algorithmic discovery.

The demographic dimensions are deeply intertwined with platform monetization strategies. The ageGroup dimension, specifically following the inclusion of under-18 estimations, plays a vital role in navigating compliance with the Children's Online Privacy Protection Act (COPPA). Viewership heavily skewed towards the AGE_13_17 or younger brackets typically yields significantly lower CPMs due to regulatory restrictions on targeted advertising and the generally lower disposable income of the demographic. Conversely, audiences peaking in the AGE_25_34 and AGE_35_44 brackets command premium ad rates, particularly in finance, technology, and real estate sectors.

Geographic Dimensions

Geographic data informs localization strategies, sponsorship negotiations, and subtitle prioritization. Geography is the single strongest determinant of advertising revenue on the platform.

Common Name	API Formatted Title	Description, Calculation, and Differentiation
Country	country / country_code	**Description:** A two-letter ISO-3166-1 country code identifying the viewer's location (e.g., US, FR, IN). Unidentified traffic is classified as ZZ.[1] **Differentiation:** The foundational geographic metric. It separates high-CPM Tier 1 nations from lower-CPM developing markets, explaining discrepancies between high view counts and low revenue yields.[1]
Province / State	province / province_code	**Description:** Identifies specific U.S. states or territories using ISO 3166-2 codes (e.g., US-TX, US-CA). Unidentified states are marked US-ZZ.[1] **Differentiation:** Provides extreme granularity for

		campaigns localized within the United States. It does not currently support subdivisions of other countries or U.S. outlying areas with their own primary country codes.[1]

The disparity in regional advertising economies means that an analyst evaluating global channel performance cannot rely on raw view counts alone. A video generating one million views predominantly from the country_code of US (United States) or GB (Great Britain) will yield exponentially higher estimated_partner_revenue than a video generating five million views from the IN (India) or PH (Philippines) country_code. Consequently, creators utilizing the province_code dimension can negotiate highly lucrative, hyper-local sponsorships (such as regional automotive dealerships or political campaigns) by proving density in specific U.S. states.

Playback Location, Traffic Sources, and Device Dimensions

Understanding exactly where a video was played, how the user navigated to it, and what hardware they utilized is arguably the most critical component of algorithmic reverse-engineering.

Common Name	**API Formatted Title**	**Description, Calculation, and Differentiation**
Playback Location Type	playbackLocationType / insightPlaybackLocationType	**Description:** Categorizes the digital real estate where the video player was instantiated.[1] Values include WATCH (standard YouTube page), EMBEDDED (third-party sites), CHANNEL (creator storefront), BROWSE (home feed), SEARCH, and SHORTS.[1] **Differentiation:** Distinguishes between native platform consumption and external syndication. High EMBEDDED traffic indicates strong external SEO, whereas high BROWSE traffic indicates strong algorithmic favorability.[1]
Playback Location Detail	playbackLocationDetail / insightPlaybackLocationDetail	**Description:** Provides the specific URL or application identifier for embedded playbacks. It is only populated when the playbackLocationType is designated as EMBEDDED.[1] **Differentiation:** Acts as a targeted drill-down. If a video goes viral off-platform, this dimension identifies the exact news article, blog, or application driving the surge.[1]
Traffic Source Type	trafficSourceType / insightTrafficSourceType	**Description:** Defines the referrer—the psychological intent that led the user to the playback location.[1] Values include ADVERTISING, YT_SEARCH, RELATED_VIDEO (suggested videos), EXT_URL, NOTIFICATION, and VIDEO_REMIXES.[1] **Differentiation:** While Playback Location tells you *where* the video played, Traffic Source tells you *how* the user arrived. It maps the viewer's journey prior to the view event.[1]
Traffic Source Detail	trafficSourceDetail / insightTrafficSourceDetail	**Description:** Provides granular details based on the parent source type.[1] **Differentiation:** If the source is YT_SEARCH, this detail provides the exact typed search query. If the source is RELATED_VIDEO, it provides the video_id of the referring video. If the source is EXT_URL, it provides the external domain.[1]
Device Type	deviceType / device_type	**Description:** Categorizes the physical hardware: Computer (101), TV (102), Game Console (103), Mobile Phone (104), or Tablet (105).[1] **Differentiation:** Hardware fundamentally alters consumer behavior. Device type correlates heavily with retention and interaction rates.[1]
Operating System	operatingSystem / operating_system	**Description:** Identifies the software architecture running the device (e.g., Windows, Android, iOS, ChromeOS, Chromecast).[1] **Differentiation:** Useful for application developers running install campaigns to determine if their audience leans heavily towards specific ecosystems.[2]
YouTube Product	youtubeProduct	**Description:** Identifies the specific YouTube service interface utilized: CORE (main site/app), GAMING, KIDS, or MUSIC.[1] **Differentiation:** Allows differentiation between standard viewership and highly specialized environments where algorithms and ad policies differ drastically (e.g., YouTube Kids).[1]
Sharing Service	sharingService / sharing_service	**Description:** Identifies the specific external platform (e.g., Twitter, Facebook, WhatsApp) utilized when a viewer clicks the native "Share" button.[1] **Differentiation:** Maps the vectors of "dark social" distribution, highlighting which external networks harbor the creator's most vocal advocates.[2]
Elapsed Video Time Ratio	elapsedVideoTimeRatio / elapsed_video_time_percentage	**Description:** Represents a specific percentile timestamp within a video's total duration.[14] **Differentiation:** This dimension is uniquely utilized in audience retention reports. It acts as the X-axis mapping coordinates against the audience_retention_percen

		tage metric, allowing analysts to construct the visual retention curve.[14]
Audience Type	audienceType / audience_retention_type	**Description:** Used in conjunction with retention metrics to classify the segment of the audience being analyzed.[14] **Differentiation:** Filters retention data to compare organic viewers against viewers acquired through paid advertising campaigns.[14]
Live or On-Demand	liveOrOnDemand / live_or_on_demand	**Description:** Segregates metrics generated during a live broadcast (LIVE) versus recorded playback (ON_DEMAND).[1] **Differentiation:** Live viewers exhibit significantly higher concurrent engagement (chat participation) but generally yield shorter overall average percentages viewed compared to highly edited, condensed VOD content.[1]
Ad Type	adType / ad_type	**Description:** Provides insights into the specific format of advertising served, distinguishing between skippable in-stream ads, non-skippable ads, bumper ads, and display ads.[2] **Differentiation:** Crucial for assessing how aggressive monetization formats impact viewer drop-off rates. High volumes of non-skippable ad_type impressions may correlate with negative spikes in audience retention.[2]

The intersection of traffic_source_type and device_type reveals profound behavioral trends. For instance, traffic originating from RELATED_VIDEO (suggested videos) often signifies that the algorithm is actively pairing the content with high-performing adjacent media. If this traffic is overwhelmingly mapped to the TV (102) or Game console (103) device_type, analysts can infer a "lean-back" viewing environment. In these environments, raw view counts and average view durations inflate massively due to prolonged, passive consumption sessions, but interactivity (clicks on cards, comments, and likes) plummets because viewers lack a mouse or touchscreen. Conversely, YT_SEARCH traffic mapping to Mobile phone (104) indicates active, high-intent problem-solving behavior, usually resulting in shorter sessions but high interactivity and sharing rates.

3. Channel-Wide Statistics and Aggregated Metrics

Channel-level statistics aggregate performance data across the entire repository of a creator's or content owner's videos. These metrics provide a macro-level view of brand health, audience acquisition, and overall monetization efficacy.

Channel Growth and Audience Acquisition

Audience acquisition metrics serve as the primary indicators of a channel's ongoing vitality and community resonance.

Common Name	API Formatted Title	Description, Calculation, and Differentiation
Total Subscribers	channel_subscribers	**Description:** The aggregate number of users currently subscribed to the channel. In public Data API outputs, this value is intentionally rounded down to three significant figures to prevent precise real-time tracking.[18] **Differentiation:** Represents the cumulative historical acquisition of the channel. It is a static vanity metric, significantly less indicative of current algorithmic momentum than the velocity of subscriber change.[18]
Total Videos	channel_videos	**Description:** The absolute count of public videos uploaded to the channel over its lifetime.[18] **Differentiation:** Used primarily as a denominator to calculate output velocity and per-video average returns. It explicitly excludes private, unlisted, and deleted content.[18]
Subscribers Gained	subscribersGained / subscribers_gained	**Description:** The total number of users who subscribed to the channel during the queried time period.[8] **Differentiation:** When mapped against individual videos, this metric reveals which specific content formats function as the most potent "top-of-funnel" acquisition tools. A high views-to-subscribers-gained ratio indicates highly resonant content.[8]
Subscribers Lost	subscribersLost / subscribers_lost	**Description:** The total number of users who actively unsubscribed during the period, or accounts that were purged by YouTube for being inactive or bots.[8] **Differentiation:** Spikes in this metric provide immediate negative feedback. They often correlate directly with drastic shifts in content strategy, highly controversial uploads, or prolonged periods of inactivity that prompt users to clean up their subscription feeds.[8]
Viewer Percentage	viewerPercentage / views_percentage	**Description:** The percentage of viewers who were actively logged into a Google/YouTube account while watching the content.[8] **Differentiation:** High logged-in viewership is an operational prerequisite for high engagement. Anonymous viewers cannot subscribe, like, or comment. A channel suffering from low engagement despite high views must evaluate this

		metric to determine if the traffic is simply unauthenticated (e.g., embedded on external sites without Google SSO).[8]
Cancellation Survey Responses	membershipsCancellationSurveyResponses	**Description:** The number of exit surveys completed by users actively canceling their paid channel memberships.[9] **Differentiation:** A highly specific, qualitative data point translated into a quantitative metric. It serves as an early warning system for creator burnout, declining perk value, or economic friction within the core community.[9]

Channel-Wide Financial and Ad Performance Metrics

Monetization metrics fluctuate wildly based on geography, audience demographics, and the nature of the content (advertiser-friendly versus restricted). All revenue metrics delivered by the API are explicitly marked as "estimated" and are subject to stringent end-of-month reconciliation adjustments to account for invalid traffic and ad fraud.[8]

Common Name	API Formatted Title	Description, Calculation, and Differentiation
Estimated Revenue	estimatedRevenue / estimated_partner_revenue	**Description:** The total estimated net revenue from *all* sources, combining Google-sold advertising, subscriptions, and non-advertising sources like Super Chats and Channel Memberships.[8] **Differentiation:** The ultimate baseline for channel profitability, representing the total holistic financial yield deposited to the creator.[8]
Ad Revenue	estimatedAdRevenue / estimated_partner_ad_revenue	**Description:** Total estimated net revenue specifically derived from Google-sold advertising sources.[9] **Differentiation:** Separates core algorithmic ad performance from direct-to-consumer monetization alternatives, allowing creators to assess their reliance on ad-market volatility.[9]
Auction Ad Revenue	estimated_partner_ad_auction_revenue	**Description:** Earnings derived exclusively from auction-sold advertising via the AdSense network.[9] **Differentiation:** This metric (available in the Reporting API) isolates revenue generated by automated, programmatic bidding, which fluctuates constantly based on real-time advertiser demand.[9]
Reserved Ad Revenue	estimated_partner_ad_reserved_revenue	**Description:** Earnings from reserved-sold advertising, typically brokered via DoubleClick and direct partner sources.[9] **Differentiation:** Represents premium, guaranteed ad placements. High reserved revenue indicates strong, direct brand interest in the channel's specific audience demographic, independent of standard algorithmic bidding.[9]
Transaction Revenue	estimated_partner_transaction_revenue	**Description:** Revenue generated from direct viewer transactions, such as paid content purchases, Super Chats, and Fan Funding, calculated after refunds have been deducted.[9] **Differentiation:** Isolates the financial power of parasocial relationships. Channels with low views but deep community loyalty will see this metric vastly outperform their standard ad revenue.[9]
Premium Revenue	estimatedRedPartnerRevenue / estimated_partner_red_revenue	**Description:** Estimated revenue generated strictly by YouTube Premium (formerly YouTube Red) subscribers watching the content.[9] **Differentiation:** A highly stable revenue source completely insulated from ad-market fluctuations, ad-blockers, and advertiser boycotts. Revenue is calculated based on the watch time proportion of the subscription pool.[9]
Gross Revenue	grossRevenue / estimated_youtube_ad_revenue	**Description:** The total estimated gross revenue from all Google-sold or DoubleClick-partner-sold ads *before* the platform's revenue split is applied.[9] **Differentiation:** Allows content owners and multi-channel networks to audit the exact percentage of the revenue share being retained by the YouTube platform.[9]
Ad Impressions	adImpressions / ad_impressions	**Description:** The verified, absolute number of times an advertisement was successfully served and viewed by a user.[9] **Differentiation:** When compared directly against total video views, this determines the "monetized playback rate," highlighting what percentage of the audience is utilizing ad-blockers or residing in non-monetized geographic regions.[9]
CPM (Cost Per Mille)	cpm / estimated_cpm	**Description:** The estimated gross revenue generated per one thousand ad impressions.[9]

		Differentiation: The ultimate indicator of advertiser demand for a specific audience. Finance and technology channels routinely secure high CPMs due to lucrative conversion funnels, whereas broad entertainment or gaming channels typically suffer lower CPMs due to younger audiences.[9]
Playback-Based CPM	playbackBasedCpm / estimated_playback_based_cpm	**Description:** Estimated gross revenue per one thousand monetized video playbacks.[9] **Differentiation:** Distinct from raw CPM. A single video playback can serve multiple discrete ad impressions (e.g., a pre-roll ad followed by a mid-roll ad). Playback-based CPM measures the yield of the entire viewing session, rather than the individual ad units.[9]
Monetized Playbacks	monetizedPlaybacks / estimated_monetized_playbacks	**Description:** Instances where a viewer initiated video playback and was served at least one ad.[9] **Differentiation:** Tracks the sheer volume of revenue-generating events, effectively filtering out views that did not trigger an ad auction or were suppressed by ad-blockers.[9]

The systemic interplay between these financial metrics yields profound strategic insights. For example, if an analyst observes a channel demonstrating a rising estimated_cpm but a simultaneously falling estimated_partner_revenue, the data paints a highly specific picture. It suggests that while the audience's underlying advertiser value is increasing (perhaps aging into a more lucrative demographic), the actual volumetric delivery of ad_impressions is plummeting. This is likely due to either a steep decline in raw total views, a massive increase in desktop ad-blocker usage, or a geographical shift in audience traffic toward non-monetized regions.

4. Individual Video Statistics: Universal Metrics

Universal metrics apply indiscriminately to all video formats on the platform, whether they are standard horizontal uploads, archived live streams, or vertical Shorts. These metrics form the bedrock of the algorithm's recommendation engine, utilizing quantitative feedback to rank content relative to its peer group.

View and Reach Metrics

Reach metrics measure the extent of a video's exposure, while view metrics quantify the conversion of that exposure into actual consumption.

Common Name	API Formatted Title	Description, Calculation, and Differentiation
Total Views	views / views	**Description:** The raw count of video views. In March 2025/2026 API updates specifically regarding Shorts, this metric was redefined to count the moment a Short simply starts to play or replay, removing previous minimum watch-time requirements.[4] **Differentiation:** The standard, public-facing unit of reach. However, due to autoplay features in Browse and the immediate scroll mechanics of Shorts, it has become increasingly decoupled from actual human "attention," serving more as a metric of exposure.[17]
Engaged Views	engagedViews / engaged_views	**Description:** The absolute number of times a video has been viewed past the initial few seconds.[8] **Differentiation:** Following the 2025/2026 API viewcount updates, engaged_views became the critical metric for measuring actual human retention. It filters out instantaneous algorithmic scrolls and autoplay misfires, revealing the true audience that decided to consume the media.[8]
Premium Views	redViews / red_views	**Description:** Views generated exclusively by authenticated YouTube Premium members.[8] **Differentiation:** Premium views bypass the advertiser auction entirely. These views always generate revenue via the distributed subscription pool, regardless of the viewer's demographic CPM or regional ad market.[9]

Engagement Metrics

Engagement data provides the qualitative feedback loop required by the recommendation systems to assess user satisfaction. High engagement validates the algorithm's decision to serve the impression.

Common Name	API Formatted Title	Description, Calculation, and Differentiation
Likes	likes / likes	**Description:** The total number of positive user ratings appended to the video.[8] **Differentiation:** A primary, frictionless signal of audience satisfaction. A high likes-to-views ratio immediately signals to the algorithm that the content delivered on the promise of its title and thumbnail.[8]
Dislikes	dislikes / dislikes	**Description:** The total number of negative user ratings.[8] **Differentiation:** While hidden from the public consumer UI to prevent targeted harassment campaigns, dislikes remain fully visible and tracked in the API. They signal polarization. However, the algorithm often treats dislikes as generic "engagement," meaning highly disliked, controversial

		videos can still achieve viral distribution if their watch time remains exceptionally high.[8]
Comments	comments / comments	**Description:** The total absolute number of comments posted beneath the video.[8] **Differentiation:** Leaving a comment requires significantly more user effort and cognitive friction than clicking a like button. High comment velocity immediately following publication triggers rapid organic push in Browse features, indicating high community resonance.[8]
Shares	shares / shares	**Description:** The number of times a video was distributed via the platform's native "Share" button.[8] **Differentiation:** Shares are the strongest indicator of "external virality." They represent users actively pushing the content outside the closed algorithmic loop of YouTube and into dark social channels (messaging apps) or other social networks, acting as organic advocates for the creator.[8]

Watch Time and Audience Retention

Watch time metrics govern the platform's core overarching objective: maximizing the duration users remain on the site to serve maximum ad impressions.

Common Name	API Formatted Title	Description, Calculation, and Differentiation
Total Watch Time	estimatedMinutesWatched / watch_time_minutes	**Description:** The aggregate duration of content consumed by all users, measured in estimated minutes.[8] **Differentiation:** This is the single most heavily weighted metric for establishing total channel authority. It is the absolute gatekeeper metric for YouTube Partner Program monetization eligibility, which requires specific watch hour thresholds.[9]
Premium Watch Time	estimatedRedMinutesWatched / red_watch_time_minutes	**Description:** The total minutes watched specifically by YouTube Premium members.[9] **Differentiation:** Directly correlates to the calculation of estimated_partner_red_revenue, as subscription payouts are prorated based on total watch time share.[9]
Average View Duration (AVD)	averageViewDuration / average_view_duration_seconds	**Description:** The average length of a video playback, expressed in absolute seconds. As of December 13, 2021, this calculation explicitly excludes traffic generated from looping clips.[8] **Differentiation:** AVD dictates the absolute volume of attention captured per click. It is critical for assessing long-form content, where retaining a viewer for 8 minutes is vastly more valuable than retaining them for 2 minutes, regardless of the video's total length.[8]
Average Percentage Viewed (APV)	averageViewPercentage / average_view_duration_percentage	**Description:** The average percentage of the video's total length watched during a playback.[8] **Differentiation:** APV normalizes watch time across different video lengths. A 50% APV on a 20-minute video is an exceptional algorithmic signal; however, a 50% APV on a 30-second Short is catastrophic. APV reveals if the content is paced correctly relative to its total runtime.[20]
Audience Watch Ratio	audienceWatchRatio / audience_retention_percentage	**Description:** The absolute ratio of viewers actively watching at a specific timestamp relative to the total number of initial views.[8] **Differentiation:** This metric, plotted against elapsed_video_time_percentage, enables analysts to construct the exact retention curve. It identifies precisely where users rewind (creating retention spikes) or abandon the video (creating retention cliffs).[8]
Relative Retention	relativeRetentionPerformance	**Description:** A normalized indexing measurement (represented as a value between 0.0 and 1.0) showing how well a video retains viewers compared to all other YouTube videos of a similar length.[8] **Differentiation:** This metric provides critical context by answering the question: "Is my audience dropping off faster or slower than the global platform average for a video of this exact duration?".[8]

Livestream Metrics

Live broadcasting introduces synchronous viewing dynamics, requiring distinct measurement parameters.

Common Name	API Formatted Title	Description, Calculation, and Differentiation
Average Concurrent Viewers	averageConcurrentViewers	**Description:** The average number of users watching

		the livestream simultaneously throughout its duration.[4] **Differentiation:** Unlike standard views which accumulate asynchronously over time, concurrent viewers measure the immediate, real-time drawing power of the creator.[8]
Peak Concurrent Viewers	peakConcurrentViewers	**Description:** The highest recorded number of simultaneous viewers at any single moment during the broadcast.[8] **Differentiation:** Identifies the climax of the stream, useful for determining when the maximum audience was present for major announcements or sponsor reads.[8]

5. Individual Video Statistics: Long-Form Only Analytics

Long-form content (traditionally horizontal video exceeding one minute, and critically, videos exceeding eight minutes which qualify for mid-roll monetization) possesses unique metadata architecture. The analytical focus here diverges from rapid exposure toward sustained narrative retention, interactive elements, and playlist continuity.[22]

Interactive Elements: Cards, End Screens, and Legacy Annotations

Interactive overlays are designed to extend the user session by funneling viewers to additional channel content, preventing them from returning to the algorithmic home feed.

Common Name	API Formatted Title	Description, Calculation, and Differentiation
Card Impressions	cardImpressions / card_impressions	**Description:** The total number of times interactive info-cards were rendered and displayed to a viewer.[9] **Differentiation:** Analyzes the baseline visibility of mid-video calls to action.[9]
Card Clicks	cardClicks / card_clicks	**Description:** The raw count of clicks registered on displayed cards.[9]
Card Click Rate	cardClickRate / card_click_rate	**Description:** The ratio of card clicks to card impressions.[9] **Differentiation:** A low click rate indicates a strategic failure: the card is either irrelevant to the current narrative moment, or it is presented at a time when the viewer is too heavily engaged with the primary video to navigate away.[9]
Teaser Impressions	cardTeaserImpressions / card_teaser_impressions	**Description:** The number of times the small "teaser" text animation appeared before the full card panel was expanded by the user.[9]
Teaser Clicks	cardTeaserClicks / card_teaser_clicks	**Description:** Clicks specifically on the teaser text overlay.[9]
Teaser Click Rate	cardTeaserClickRate / card_teaser_click_rate	**Description:** The ratio of teaser clicks to teaser impressions.[9] **Differentiation:** Measures the efficacy of the copywriting utilized in the teaser pop-up to interrupt viewing patterns.[9]
End Screen Impressions	N/A (Reporting API) / end_screen_element_impressions	**Description:** The number of times end screen elements (linked videos, playlists, or subscribe buttons) were displayed in the final 5–20 seconds of a video.[9]
End Screen Clicks	N/A (Reporting API) / end_screen_element_clicks	**Description:** Total clicks on those specific end screen elements.[9]
Annotation Impressions	annotationImpressions / annotation_impressions	**Description:** Total impressions of legacy annotations.[9] **Differentiation:** Annotations were deprecated and removed from the platform's consumer UI years ago. However, these metrics remain active as "Core Metrics" in the API solely to maintain historical data consistency for older channel archives.[8]
Annotation Clicks	annotationClicks / annotation_clicks	**Description:** Total historical clicks on annotations.[9]
Annotation CTR	annotationClickThroughRate / annotation_click_through_rate	**Description:** Ratio of clicks to total clickable annotation impressions.[9]
Annotation Closes	annotationCloses / annotation_closes	**Description:** Total historical closures of annotations.[9]
Annotation Close Rate	annotationCloseRate / annotation_close_rate	**Description:** Ratio of closures to total annotation impressions.[9]

Playlist Interactivity Metrics

Playlists are a vital mechanism for long-form content, driving binge-watching behavior and significantly inflating overall watch_time_minutes without requiring the algorithm to continuously recommend the next video.

Common Name	API Formatted Title	Description, Calculation, and Differentiation
Playlist Starts	playlistStarts / playlist_starts	**Description:** The number of times a user initiated playback of a full playlist, rather than a standalone video (measured on web platforms).[9] **Differentiation:** Represents a distinct user intent to consume bulk, serialized content rather than casual single-video browsing.[9]
Playlist Views	playlistViews / views (aggregated)	**Description:** The number of times videos were viewed strictly within the context of a playlist.[8] **Differentiation:** Isolates views generated by curated sequencing from organic views generated by

		standard algorithmic distribution.[8]
Views Per Playlist Start	viewsPerPlaylistStart	**Description:** The average number of videos watched per playlist initiation. **Differentiation:** A critical metric for assessing playlist cohesion. If this number sits near 1.0, the playlist is failing its intended purpose of driving sequential views; if it is >3.0, the playlist effectively traps user attention and drives massive aggregate watch time.
Average Time in Playlist	averageTimeInPlaylist	**Description:** The average estimated minutes a viewer spends consuming content within a single playlist session. **Differentiation:** Measures the total durational yield of the playlist sequence, combining the AVDs of all individual videos watched during the start event.
Playlist Saves	playlistSaves / playlist_saves_added / playlist_saves_removed	**Description:** The absolute number of times users saved (or removed) a playlist to their personal library.[9] **Differentiation:** Indicates extraordinarily high perceived value and a strong intent to return for future viewing sessions, acting as a powerful positive algorithmic signal.[9]
Videos Added to Playlist	videosAddedToPlaylists / videos_added_to_playlists	**Description:** The number of times a specific video was added to any user-generated playlist (explicitly excluding automatic "watch later" or history playlists).[9] **Differentiation:** Acts as a proxy for "favorite" status, guaranteeing future recurring views and validating the video's long-term utility.[9]
Videos Removed	videosRemovedFromPlaylists / videos_removed_from_playlists	**Description:** The absolute number of times videos were removed from any user playlist.[9]

The Long-Form Algorithmic Imperative: CTR vs. AVD

In long-form analytics, the algorithmic success of a video is almost entirely dictated by the relationship between two metrics: the Click-Through Rate and the Average View Duration (AVD).[21]

The API exposes the click-through rate via videoThumbnailImpressionsClickRate. A high CTR (e.g., >8%) combined with a low AVD explicitly indicates to the algorithm that the thumbnail and title are acting as clickbait—overpromising the premise and underdelivering on the execution. The algorithm detects this psychological mismatch via rapid user abandonment and subsequently throttles the video's impressions to protect the user experience.[21]

Conversely, a video demonstrating a moderate CTR (4-5%) but a sustained AVD (maintaining >50% APV) demonstrates high audience satisfaction. This prompts the recommendation engine to continually expand the video's reach to broader, adjacent audience clusters.[21] Furthermore, only long-form videos contribute to the primary Watch Hour requirements for standard YouTube Partner Program monetization.[19] While short-form content generates rapid reach, long-form content drives deeper psychological engagement and vastly higher RPMs (Revenue Per Mille) due to the insertion of mid-roll advertising, which strictly requires videos to exceed the 8-minute duration threshold.[22]

6. Individual Video Statistics: Short-Form Only Analytics

YouTube Shorts represent a fundamental paradigm shift in video delivery, utilizing a vertical, infinite-scroll interface. Consequently, the metrics required to analyze Shorts deviate sharply from long-form standards. In the Shorts ecosystem, the concept of a "click" is largely eliminated, replaced entirely by the immediate "swipe."

The Anatomy of a Short-Form View

Common Name	API Formatted Title	Description, Calculation, and Differentiation
Shorts View	views / views	**Description:** Following the comprehensive API updates of March 2025/2026, a view on a Short is counted the precise moment the media starts to play or replay, removing any previous minimum watch-time requirements.[4] **Differentiation:** This redefinition drastically inflates raw view counts, aligning YouTube's statistical reporting with competitors like TikTok and Instagram Reels.[23] It transforms the "view" from a metric of intent into a metric of mere exposure.[17]
Engaged View	engagedViews / engaged_views	**Description:** Views that successfully bypass the immediate scroll and persist past the initial few seconds of playback.[8] **Differentiation:** Because the views metric is now triggered instantaneously upon rendering, engagedViews has become the true, critical measure of a Short's ability to hook an audience. It mathematically filters out users who passively swiped past the content without registering it.[17]
Shown in Feed	videoThumbnailImpressions / video_thumbnail_impressions	**Description:** The number of times a Short was physically rendered and presented in a user's infinite Shorts feed.[8] **Differentiation:** In the context of Shorts, a "thumbnail impression" is effectively an autoplay event in the feed, not a static image waiting to be clicked. If this metric is exceptionally high but engagedViews is low, the opening frame of the video failed completely to arrest the user's scrolling momentum.[8]
Swiped Away	N/A (Derived)	**Description:** Calculated analytically by contrasting "Shown in Feed" (videoThumbnailImpressions) with engagedViews.[24] **Differentiation:** The

		ultimate negative signal for short-form content. If viewers consistently swipe away within the first second, the algorithm immediately learns the content is unappealing and terminates the video's reach in the feed.[25]

Short-Form Algorithmic Benchmarks: The Density Engine

Because Shorts range strictly from 15 to 60 seconds, the retention benchmarks are fundamentally altered. A 50% Average Percentage Viewed (APV), which would be considered excellent for a 10-minute long-form video, is a "death sentence" in the algorithmic Shorts feed.[20]

The short-form algorithm demands extreme narrative density. To achieve organic viral velocity, a 60-second Short typically requires an APV greater than 90%, while a 15-second Short requires an APV exceeding 130%.[20] An APV crossing the 100% threshold indicates that the average user watched the entire video and allowed the player to loop, generating multiple views per session. This looping behavior is the most critical behavioral signal for vertical video distribution.[20]

Remixes, Audio Pivots, and Derivative Traffic

Short-form content thrives on derivative creation and viral audio trends. The APIs provide specific dimensions to track how original content spawns secondary media.

Common Name	API Formatted Title	Description, Calculation, and Differentiation
Remix Traffic	trafficSourceType = VIDEO_REMIXES	**Description:** Identifies views originating specifically from the "remixed video" attribution link displayed inside the Shorts player.[1] **Differentiation:** When a creator's original audio or visual snippet is sampled and used by another creator, this dimension tracks the downstream traffic flowing backward to the original source, quantifying the value of starting a trend.[1]
Sound Page Traffic	playbackLocationType = SOUND_PAGE	**Description:** Views originating from the dedicated audio pivot page that aggregates all Shorts utilizing a specific sound or song.[26] **Differentiation:** Highly relevant for music labels, recording artists, and trend-based creators tracking the viral spread and adoption rate of an audio meme.[26]

While Shorts are unparalleled for rapid discovery, rapid content testing, and top-of-funnel reach, their direct monetization potential is fractionally comparable to long-form content.[22] Ad revenue generated from the Shorts feed is pooled and distributed based on aggregate view share, resulting in RPMs that are often measured in pennies per thousand views, compared to several dollars per thousand views for long-form mid-rolls.[22]

7. External Analytics: GA4 and Ecosystem Integrations

YouTube's native APIs explicitly track telemetry *on* the platform (YouTube.com and official apps). However, a holistic data strategy requires understanding how embedded video assets perform across the broader internet. This is achieved via Google Analytics 4 (GA4) integrations.

When enhanced measurement is enabled in GA4, or when custom tags are deployed via Google Tag Manager, analysts can track the performance of embedded YouTube players on their proprietary websites.[11] GA4 utilizes event-based tracking rather than session-based tracking, categorizing video interactions into distinct events:

- video_start: Fired the moment the embedded video begins playing.[13]
- video_progress: Fired at specific percentage milestones (e.g., 10%, 25%, 50%, 75%) to track external retention.[13]
- video_complete: Fired when the video reaches its conclusion.[13]

By creating custom metrics within GA4—such as assigning a binary 1 to a video_view parameter triggered specifically by the video_start event—analysts can quantify external engagement.[13] Furthermore, by intersecting these custom video metrics with standard GA4 dimensions like session_source (which tracks where the user came from, e.g., "google", "facebook", "newsletter"), analysts can determine which external acquisition channels drive the highest video engagement on their landing pages.[12]

This integration effectively closes the data loop. A user might discover a brand via a high-APV Short (tracked via native API engagedViews), actively search for a deeper dive (tracked via native YT_SEARCH traffic source), and finally click an external link to purchase a product on the brand's website where they watch a final product demo (tracked via GA4's video_complete and standard UTM tracking).[11]

8. Strategic Synthesis

The true value of this exhaustive API architecture lies not in the observation of isolated metrics, but in the correlative synthesis of overlapping dimensions.

The API landscape clearly bifurcates content strategy. Long-form video operates as a "Commitment Engine," relying on the friction of the click (videoThumbnailImpressionsClickRate) and the sustained psychological investment of watch time (averageViewDuration).[21] Success here yields deep audience loyalty, sustained premium ad revenue, and high subscriber conversion rates.[19] Conversely, Short-form video operates as a "Density Engine." The click is irrelevant; the metric of entry is exposure (videoThumbnailImpressions).[24] Success demands looping retention profiles exceeding 100% APV, yielding massive instantaneous reach but significantly lower immediate financial returns.[20]

Ultimately, the Google and YouTube analytics systems present a highly structured, multidimensional topography. From the macro-level revenue aggregations of the Reporting API, down to the granular, millisecond-level retention curves of the Analytics API, mastery of these schemas allows for the precise, empirical reverse-engineering of user behavior, algorithmic preference, and digital monetization.

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